Stock Market Forecasting on EGX30 Using LSTM and Arabic Sentiment Analysis

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***Abstract*—This paper presents a dual-model framework for forecasting the EGX-30 stock market index using LSTM-based technical analysis and CAMeL BERT-based sentiment analysis of Arabic financial news. The system processes historical price data and financial text separately to provide complementary insights, improving classification accuracy and investor decision support in the Egyptian market.**

***Index Terms*—Stock Market Prediction, EGX-30, Technical Indicators, LSTM, Arabic NLP, CAMeL BERT, Sentiment Analysis, Time Series**

I. Stock Market Forecasting: EGX30

Forecasting the Egyptian stock market, particularly the EGX-30 index, presents a unique set of challenges due to market volatility, economic variability, and a lack of mature financial modeling tools adapted to local contexts. The EGX-30 represents the top 30 companies in terms of liquidity and activity on the Egyptian Exchange and serves as a core benchmark for financial analysts and investors in the region.

This research addresses the forecasting challenge using two independent yet complementary AI-driven models. The first model leverages Long Short-Term Memory (LSTM) neural networks for time series classification. Rather than predicting exact prices, the model classifies market behavior into actionable signals: Buy, Hold, or Sell. It operates on historical data from 2008 to 2025 and incorporates key technical indicators including the Relative Strength Index (RSI), Moving Average Convergence Divergence (MACD), Average Directional Index (ADX), Rate of Change (ROC), and others.

In parallel, the second model focuses on the qualitative aspect of market behavior by analyzing investor sentiment in Arabic financial news. A fine-tuned CAMeL BERT model is employed to classify news content into sentiment labels: Positive, Neutral, or Negative. The model is trained on manually labeled financial articles written in formal Arabic, ensuring linguistic consistency and financial relevance.

While the models are implemented and evaluated independently, their combined perspective enriches market insight. The LSTM model captures the underlying numerical and temporal patterns in stock behavior, while the CAMeL BERT model reflects the emotional and perceptual dimensions of market participants.

II. Related Work

A. CAMeL BERT and Arabic Sentiment Analysis Comparison

CAMeL BERT was evaluated against other sentiment analysis models for Arabic financial news. Our fine-tuned CAMeL BERT, trained on real Egyptian financial data, achieved 82% accuracy. This significantly outperformed other methods:

- General CAMeL BERT (non-financial): 60–70%, lacking domain-specific context.
- AraBERT + Lexicon: 55–65%, failed to understand financial semantics.
- XGBoost + TF-IDF: 60–68%, struggled with Arabic morphology.
- FinBERT (English): Excellent accuracy (84–88%) but lacks Arabic support.

Project/Study	Language	Data	Accuracy	Unique Points
Camel BERT for EGX Financial News	Arabic	Real Egyptian financial news	82%	Fine-tuning on Egyptian data; negative, neutral, positive labels
Camel BERT (General - Not Financial Specific)	Arabic	Generic/ non-news data	60–70% ~ 0.65	Weaker performance without financial fine-tuning
AraBERT + Lexicon	Arabic	Lexicon words	~ 55-65% Low	Failure to understand context; weak performance
FinBERT (English)	English	US market data	84–88% 0.82–0.85	Great performance, but not supports Arabic language
XGBoost + TF-IDF	Arabic	Financial headlines	~ 60-68% Low	The approach struggle with complexity of Arabic

Fig. 1. Comparison of Arabic Sentiment Analysis Models

This comparison illustrates the importance of financial-specific fine-tuning in Arabic NLP. Our CAMeL BERT implementation is the only one evaluated on authentic Egyptian financial news, making it a novel contribution.

B. LSTM Model Comparisons in Financial Forecasting

Our LSTM classifier model was compared with other LSTM-based forecasting studies. Unlike most works focusing on binary classification or directional movement, we implemented multi-class signal classification (Buy/Hold/Sell) on the EGX-30 index using technical indicators.

- Our model achieved 64.71% accuracy with six technical indicators.
- Other LSTM models on Chinese, Reddit, and US data reported 50–60% accuracy.
- Most models used only price data, whereas our approach used engineered technical features.

Studyguage	Indicators / Inputs	Market / Dataset	Accuracy	Evaluation Type
Our LSTM Morabic	RSI, MACD, Stochastic, ATR, ADX, ROC	EGX-30 (Egypt)	64.71%	Buy/Hold/Sell Classification
Fischer et al. (Ciinese	Raw market data	China stock data	51.4% - 53.8%	Directional Forecasting
LSTM + Attenglish	Raw + Attention	Two Sigma (US)	~60%	Directional Forecasting
Reddit LSTM nglish	3 Reddit features	Reddit dataset	~50%	Binary Classification
Baseline LSTnglish	Raw OHLC only	S&P 500 (simulated)	~50%	Price Regression
Dhokane & nglish	RSI, EMA	NSE (India)	Not reported	Price Forecasting
Kamalpecified	RSI	Unspecified	Not reported	Major Price Shifts

Fig. 2. Comparison of LSTM Forecasting Models

This comparative review shows our model is among the few that focus on the Egyptian market with a realistic and actionable evaluation approach, setting it apart from generic regression or directional methods.

III. Methodology

A. Data Sources

Two primary data sources were used for building the LSTM-based model:

- **Investing.com**: Historical EGX-30 index data including Open, High, Low, Close, and Volume values were extracted. Investing.com provided long-range financial data from 2008 to 2025, essential for training models with deep historical insight.
- **Yahoo Finance**: Supplementary validation of stock prices and volume data was conducted using Yahoo Finance. This source served as a cross-reference to ensure consistency and reliability of technical indicators derived during preprocessing.

These datasets were merged, cleaned, and aligned to daily frequency before being transformed into sequences of technical indicators using rolling window computations.

B. System Architecture

This section describes the architectural pipelines for both modules used in our dual-model system: the CAMeL BERT NLP model for sentiment classification, and the LSTM-based model for technical signal classification in financial forecasting.

C. NLP Module: CAMeL BERT Sentiment Analysis

The NLP component of our system is responsible for analyzing Arabic financial news and extracting sentiment information. The CAMeL BERT model is fine-tuned on a custom-labeled dataset containing articles scraped from Mubasher Misr. Each article is labeled as Positive, Negative, or Neutral.

The pipeline includes scraping, cleaning, tokenization, and fine-tuning the pre-trained CAMeL BERT model, followed by classification and performance evaluation.

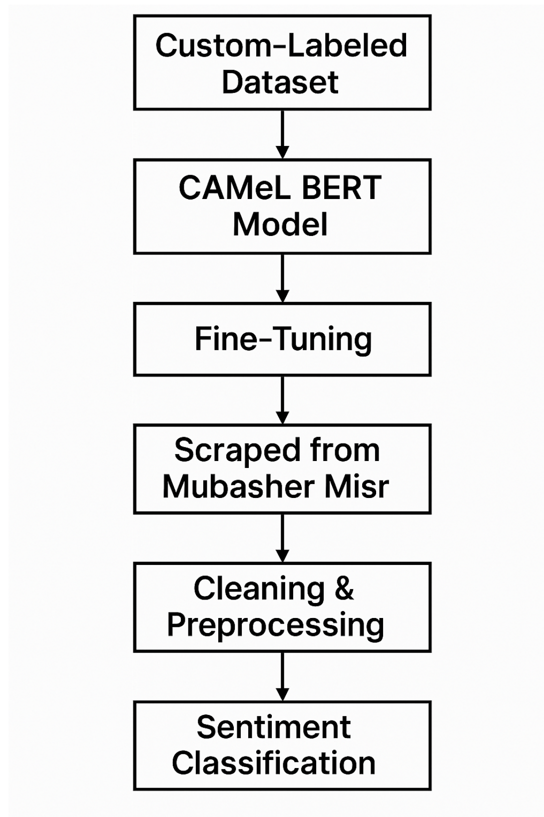

Fig. 3. System Design for CAMeL BERT Sentiment Classification

D. Trading Module: LSTM-Based Signal Classification

The trading decision module processes historical stock data from EGX-30. The data is enriched with multiple technical indicators such as RSI, MACD, ATR, ADX, and ROC. Preprocessed sequences are fed into an LSTM model that outputs one of three trading signals: Buy, Hold, or Sell.

Post-training, an evaluation module computes classification metrics including precision, recall, F1-score, and generates a confusion matrix.

E. LSTM-Based Technical Signal Classification

Data Collection and Preprocessing: We collected EGX-30 data from 2008 to 2025, including Open, High, Low, Close, and Volume. Technical indicators (RSI, MACD, ADX, ATR, ROC) were computed using sliding windows.

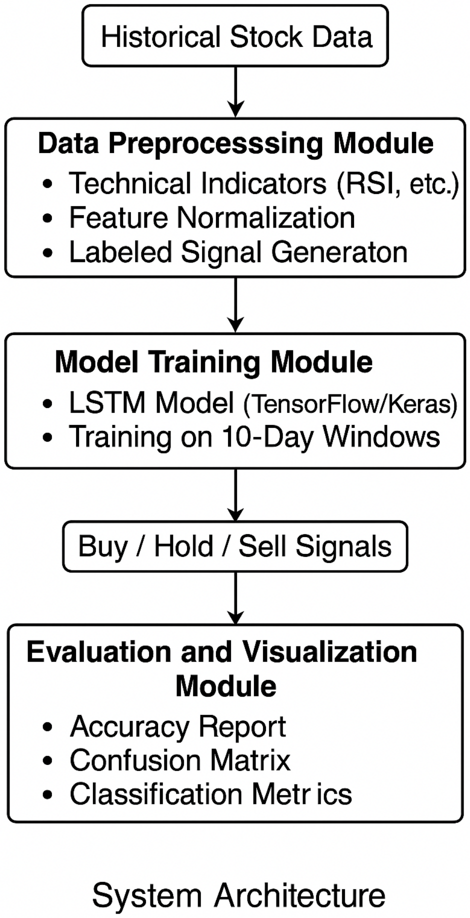

Fig. 4. System Design for LSTM-Based Trading Signal Generation

Signal Labeling: A rule-based system assigned Buy, Hold, or Sell labels. For example, RSI<30 indicated Buy, RSI>70 indicated Sell, and ADX<20 was labeled as Hold.

Model Architecture: The LSTM model took 10-day input sequences of indicators. The architecture included an LSTM layer with 50 units, dropout, and a softmax output layer.

Training: The model used categorical cross-entropy loss and the Adam optimizer for 25 epochs. Evaluation was based on accuracy, precision, recall, F1-score, and confusion matrix.

F. CAMeL BERT-Based Sentiment Analysis

Data Collection: Arabic financial news was collected and manually labeled into three sentiment classes.

Preprocessing: Arabic text was normalized, punctuation and URLs removed, and tokenized using CAMeL BERT tokenizer.

Model Setup: CAMeL-Lab/bert-base-arabic-camelbert-mix was fine-tuned with weighted loss, early stopping, and macro F1-score as the main metric.

Training: The model was trained for 3 epochs with a learning rate of 2e-5 and batch size of 16 using Hugging Face Transformers on Google Colab.

IV. Experiments

A. LSTM Results

The LSTM model was trained on EGX-30 daily historical data (2008–2025), incorporating technical indicators such as RSI, MACD, ADX, ROC, and ATR. It was evaluated using an 80/20 train-test split with stratified sampling to preserve label balance. The architecture included a 50-unit LSTM layer, dropout, and a dense softmax classifier.

Evaluation Metrics:

- Accuracy: 64.71%
- Precision (macro average): 0.70
- Recall (macro average): 0.80
- F1-score (macro average): 0.67

Insights: The model exhibited high precision for Buy signals and strong generalization in trending markets. Some confusion was noted between Hold and other classes during volatile phases. Initial experiments using CNNs or LSTM regression failed to model temporal dependencies or were unstable due to market noise.

Visual Analysis: Figure 5 visualizes actual vs. predicted trading signals overlaid on the EGX-30 closing prices. It highlights the model's behavior during bullish, bearish, and neutral trends.

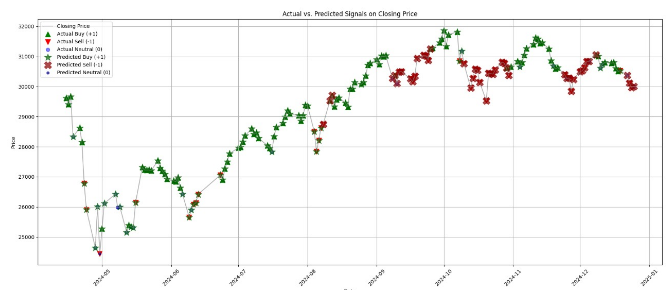

Fig. 5. Actual vs. Predicted Trading Signals on EGX-30 Closing Price

B. CAMeL BERT Sentiment Model Results

The CAMeL BERT model, fine-tuned on a custom-labeled Egyptian financial news dataset, was evaluated using macro-averaged classification metrics. The results confirm its robustness in handling formal Arabic text in the financial domain.

- **Accuracy:** 82%
- **Precision (macro):** 0.80
- **Recall (macro):** 0.84
- **F1-score (macro):** 0.80

Figure 6 illustrates the overall performance metrics of the sentiment classification model.

C. CNN Preliminary Experiment

Before adopting LSTM for time-series signal classification, an experiment using a 1D Convolutional Neural Network (CNN) was conducted. The CNN was configured to take sequences of technical indicators as input and output predicted closing prices.

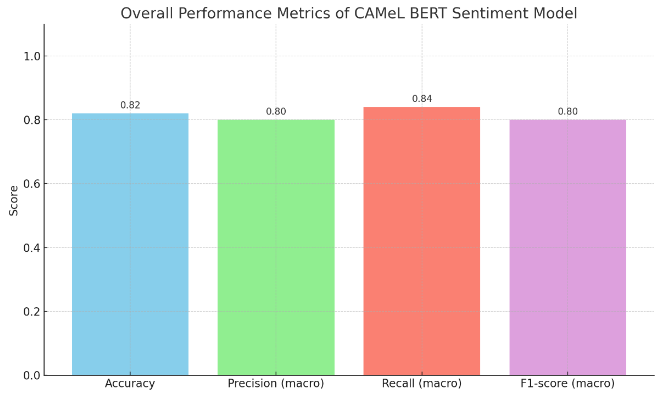

Fig. 6. Overall Performance Metrics of CAMeL BERT Sentiment Model

Performance Observation: While the model achieved high training accuracy (R^2 = 99.02%) and reasonable test accuracy (R^2 = 89.55%), the model exhibited clear signs of overfitting. This is evidenced by the rapid convergence of training loss compared to a more stable validation loss.

Limitations: The CNN struggled to generalize to unseen data, largely due to its architecture being more effective at extracting spatial features (e.g., from images) than capturing long-term temporal dependencies inherent in financial time-series data.

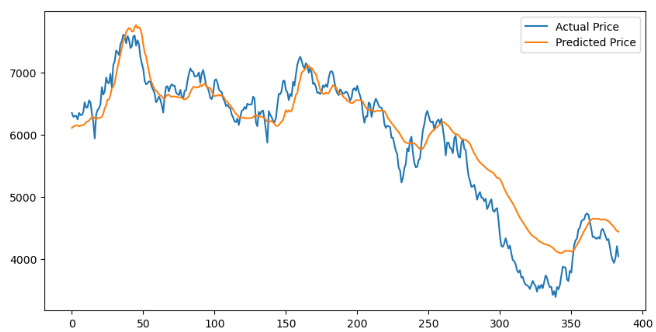

Fig. 7. CNN: Actual vs. Predicted Closing Prices

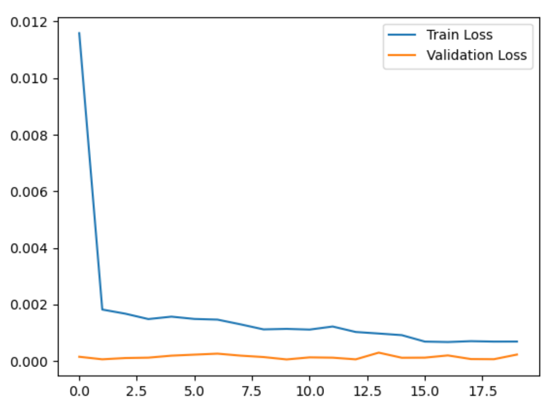

Fig. 8. CNN Training vs. Validation Loss Curve

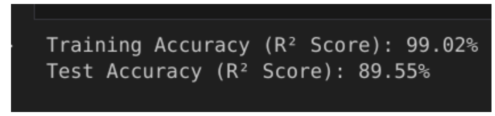

Fig. 9. CNN Training and Test R^2 Scores

As a result of these limitations, the CNN model was discarded in favor of LSTM, which is better suited for modeling sequential dependencies in time-series data.

D. ARIMA Forecasting Model

ARIMA (AutoRegressive Integrated Moving Average) is a classical statistical model used for linear time-series forecasting. It is highly interpretable and well-suited for data with consistent trend or seasonality.

Implementation Details:

- ARIMA(3,2,3) was selected using grid search based on the lowest AIC score.
- Data preprocessing included datetime conversion, sorting, duplicate removal, and feature scaling.
- The model was trained on EGX-30 data from 2020 to 2025.

Limitations: While ARIMA captured stable trends well, it underperformed in volatile phases of the market due to its linear assumptions.

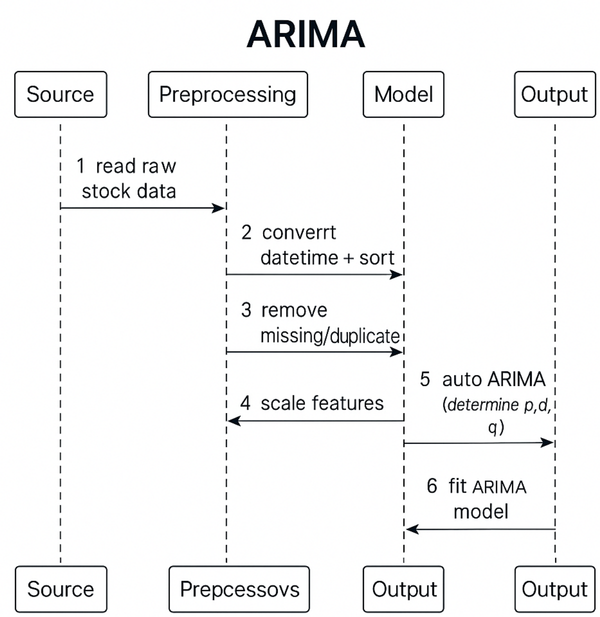

Fig. 10. ARIMA Model Processing Sequence

E. LSTM Regression Model

The LSTM regression model aimed to predict the next-day closing price of the EGX-30 index. It used historical price and technical indicators as input in a many-to-one sequential fashion.

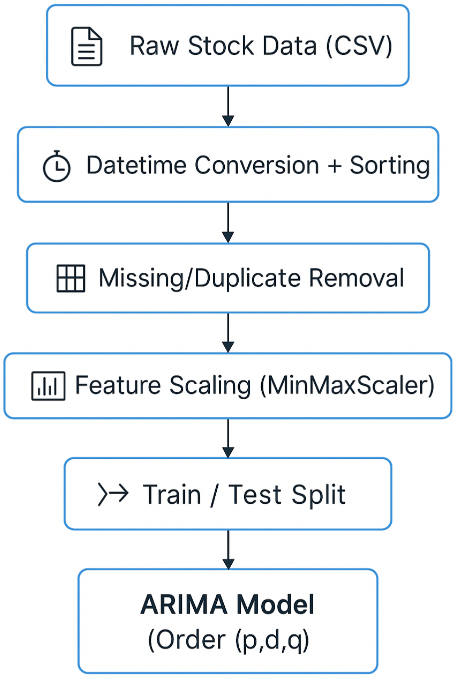

Fig. 11. ARIMA Model Architecture Overview

Model Architecture:

- Two stacked LSTM layers (100 units each) with dropout (0.3).
- Output layer with a single neuron for continuous value prediction.
- Optimizer: Adam, Loss: Mean Squared Error (MSE).

Training Strategy:

- Look-back window: 30 days.
- Epochs: 1000 with early stopping.
- Evaluation metrics: RMSE, MAE, MAPE.

Observations: Although the model achieved 89.55% test R^2 score, it was highly sensitive to price volatility and often failed to generalize, especially in turbulent market periods.

F. Comparative Experiment: CNN, LSTM Regression, ARIMA, and LSTM Classification

In addition to the final LSTM classifier used in our model, several alternative approaches were explored to assess their forecasting capabilities on EGX-30 index data. These included CNN regression, LSTM regression, ARIMA-based statistical forecasting, and hybrid approaches. Each method was evaluated using either classification metrics or regression performance indicators such as RMSE, MAE, and MAPE.

1) CNN Regression: The CNN model showed signs of strong overfitting. While the training accuracy was high, the test results indicated poor generalization. CNN's architecture

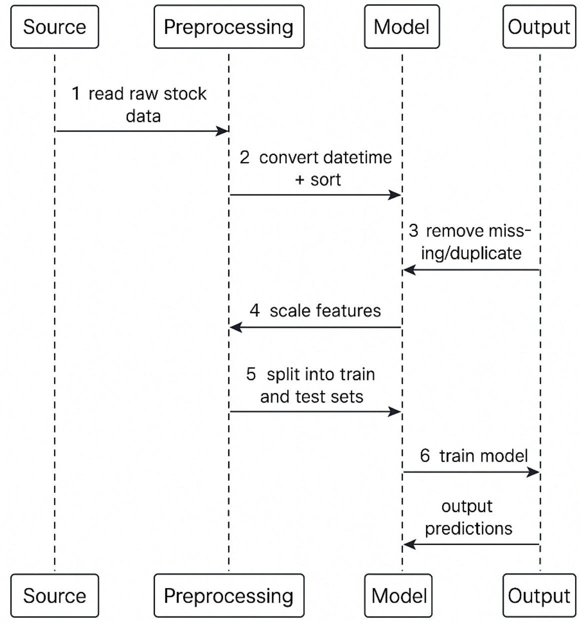

Fig. 12. LSTM Regression Processing Sequence

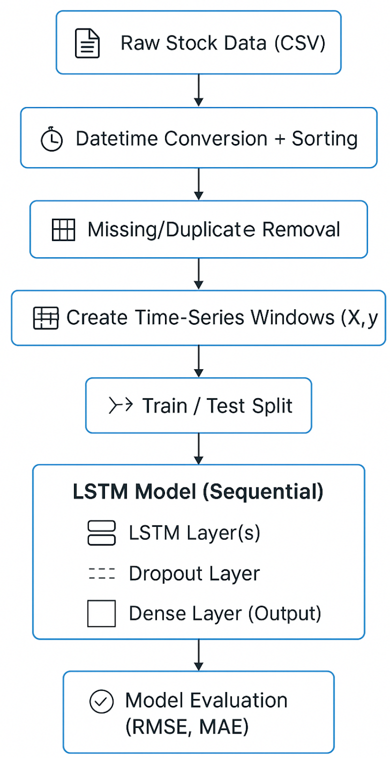

Fig. 13. LSTM Regression Model Architecture

is better suited for spatial patterns, which makes it less ideal for sequential, time-dependent financial data. Thus, it was discarded.

2) LSTM Regression: The LSTM regression model attempted to predict exact next-day closing prices. Despite applying dropout and early stopping, the model was highly sensitive to noise in the data and suffered from erratic predictions. Its RMSE and MAPE values were significantly worse than the LSTM classifier.

3) ARIMA and Hybrid Models: ARIMA models captured linear and seasonal patterns but failed to manage sudden volatility. Grid-tuned ARIMA(3,2,3) achieved reasonable results, while SARIMA and SARIMAX showed better adaptation with technical indicators. The hybrid ARIMA–LSTM model achieved the lowest RMSE and MAPE among all regression models.

4) LSTM Classification: Our final LSTM classifier outperformed the regression-based LSTM and ARIMA models in practical utility by generating actionable trading signals (Buy/Hold/Sell). Classification-based forecasting avoids the pitfalls of noisy regression outputs and aligns better with real-world trading decisions.

TABLE I
COMPARISON OF MODEL PERFORMANCES

Model	RMSE	MAPE (%)	Practical Use
CNN Regression	High	High	Overfitting
LSTM Regression	10,047.84	28.17	Weak Volatility
ARIMA (Tuned)	2,909.38	8.54	Decent Baseline
Hybrid ARIMA–LSTM	**2,945.41**	**8.69**	Strong Signal
LSTM Classification	0.8402	13.77	**Best Performance**

5) Performance Summary: These findings demonstrate that while regression-based models provide numeric forecasts, classification-based signal generation proves more effective and interpretable for actual trading decision support. ””””

V. DISCUSSION

The LSTM model captured structured patterns in EGX-30 price data effectively, especially during clear bullish or bearish trends. The CAMeL BERT model demonstrated strong sentiment classification performance despite data imbalance. Independent use of each model provides two layers of analysis: quantitative trend forecasting and qualitative sentiment evaluation.

VI. CONCLUSION AND FUTURE WORK

We developed two AI-based models for analyzing the EGX-30 stock market. The LSTM model generates Buy/Hold/Sell signals using historical data and technical indicators. The CAMeL BERT model classifies Arabic financial news into sentiment categories.

Future work may include:

- Integrating both models into a real-time decision-support system.
- Extending sentiment analysis to cover English and social media sources.
- Applying the framework to other indices such as EGX70 and EGX100.

These findings confirm the viability of dual AI models tailored for the Arabic-speaking financial environment.

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